

CSS (Cascading Style Sheets)

✓ Introduction to CSS

- CSS is used to style HTML elements.
- It controls the layout, colors, fonts, spacing, etc.
- Separates content (HTML) from presentation (CSS).

✓ Types of CSS

1. Inline CSS

- Added directly inside an HTML element using the `style` attribute.
- Example:

```
<p style="color: blue;">This is blue text.</p>
```

2. Internal CSS

- Defined inside a `<style>` tag in the `<head>` section.
- Example:

```
<style>
  p { color: red; }
</style>
```

3. External CSS

- CSS rules are written in a separate `.css` file and linked using `<link>`.
- Example:

```
<link rel="stylesheet" href="styles.css">
```

✓ CSS Selectors

- Used to select HTML elements for styling.

✓ 1. Universal Selector (*)

Purpose: Selects **all elements** on the page.

Syntax:

```
* {  
  
  margin: 0;  
  padding: 0;  
}
```

Explanation:

- This is often used to remove default spacing from all HTML elements.
 - It applies the rule to **every element** in the document.
-

✓ 2. Element Selector (`tagname`)

Purpose: Targets **specific HTML elements** by name.

Syntax:

```
p {  
  color: blue;  
}
```

Explanation:

- All `<p>` tags will be styled with blue text.
 - It only affects the specified HTML tag.
-

✓ 3. Class Selector (`.classname`)

Purpose: Selects **any element** with a specific `class`.

Syntax:

```
.btn {  
  padding: 10px;  
  background-color: lightblue;  
}
```

HTML Usage:

```
<button class="btn">Click Me</button>
```

Explanation:

- Can be reused for multiple elements.
- Good for applying **common styling**.

✓ 4. ID Selector (#idname)

Purpose: Selects **one specific element** with a unique id.

Syntax:

```
#header {  
  font-size: 24px;  
  color: white;  
}
```

HTML Usage:

```
<div id="header">Welcome</div>
```

Explanation:

- Should be **unique per page**.
- Used for targeting one particular element.

✓ 5. Group Selector (selector1, selector2)

Purpose: Applies the **same style** to **multiple selectors**.

Syntax:

```
h1, h2, h3 {  
  font-family: Arial;  
  color: navy;  
}
```

Explanation:

- Saves time and avoids repetition.
- All selected elements share the same rules.

✓ 6. Descendant Selector (parent child)

Purpose: Targets an element **inside another element**.

Syntax:

```
div p {  
  color: gray;  
}
```

HTML Example:

```
<div>  
  <p>This will be gray</p>  
</div>
```

Explanation:

- Only affects `<p>` tags **inside** a `<div>`.
- Useful for nested elements.

Session 2: Styling with CSS

1. Font Types

CSS provides several ways to style fonts using the `font-family` property.

```
p {  
  font-family: Arial, Helvetica, sans-serif;  
}
```

- **Generic families:** serif, sans-serif, monospace, cursive, fantasy
- Always provide **fallback fonts** for better compatibility.

2. Text Formatting

Used to change how text looks.

```
h1 {  
  font-size: 32px;      /* size of the font */  
  font-weight: bold;    /* normal | bold | lighter | 100 - 900 */  
  font-style: italic;   /* normal | italic | oblique */  
  text-align: center;   /* left | right | center | justify */  
}
```

```
text-decoration: underline; /* none | underline | overline | line-through */
text-transform: uppercase; /* none | capitalize | uppercase | lowercase */
line-height: 1.5; /* spacing between lines */
letter-spacing: 2px; /* spacing between letters */
word-spacing: 5px; /* spacing between words */
}
```

✓ 3. Border

Defines the edge of an element.

```
.box {
  border: 2px solid black; /* width style color */
  border-radius: 10px; /* rounded corners */
}
```

- **Styles:** solid, dashed, dotted, double, groove, ridge, etc.

✓ 4. Margin

Creates space **outside** the element.

```
.container {
  margin: 20px; /* all sides */
  margin-top: 10px;
  margin-right: 15px;
  margin-bottom: 10px;
  margin-left: 15px;
}
```

- You can also use shorthand:
margin: 10px 15px 10px 15px; → top, right, bottom, left

✓ 5. Padding

Creates space **inside** the element (between content and border).

```
.box {
  padding: 10px; /* all sides */
  padding: 5px 10px; /* top-bottom, left-right */
}
```

- Like margin, you can use top, right, bottom, left separately.

✓ 6. Background

Controls background of an element.

```
body {  
  background-color: lightblue;           /* solid color */  
  background-image: url("bg.jpg");      /* image background */  
  background-repeat: no-repeat;         /* repeat | no-repeat | repeat-x |  
repeat-y */  
  background-size: cover;               /* contain | cover | auto */  
  background-position: center;          /* position image */  
}
```

✓ 7. Display

Defines how an element is displayed on the page.

```
div {  
  display: block;                       /* default for <div> */  
}  
  
span {  
  display: inline;                     /* default for <span> */  
}  
  
.menu {  
  display: flex;                       /* flexbox layout */  
}  
  
.box {  
  display: none;                       /* hides the element */  
}
```

✓ 8. Additional Styling (Etc.)

Here are a few more useful styling properties:

✓ width and height:

```
img {  
  width: 300px;  
  height: auto;  
}
```